

Josiah St. Bernard

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EDUCATION

New York University - Tandon School of Engineering

Expected May 2028

Bachelor's Degree, Electrical and Computer Engineering

Relevant Coursework: Robotic Manipulation & Locomotion, Signals & Systems, Linear Algebra & Differential Equations, Fundamentals of Electric Circuits

Borough of Manhattan Community College

2025

Associate Degree, Computer Science (Honors)

Relevant Coursework: Data Structures, Digital Logic & State Machine Design, Discrete Mathematics

SKILLS

- **Languages:** C++, C, Python, Java, JavaScript, HTML/CSS
- **Frameworks & Tools:** CUDA, IsaacLab, IsaacSim, MuJoCo, Is, ROS, Flask, OpenCV, LangChain, Ollama, SQL, Linux, Git, Pytorch
- **Hardware & Embedded Systems:** Arduino, Raspberry Pi, Motor Drivers, PIDs, Actuators, LiDAR

EXPERIENCE

Controls Subteam Member

NYU ARC Robotics

September 2026 – Present

- Engineering for NYU's combat robotics team using ROS2, Nav2, and SLAM on NVIDIA Jetson, directly supporting the team's ARCC/RMNA competition readiness.
- Maintaining a weekly VIP engineering notebook documenting design iterations, testing results, and next steps ahead of the ARCC/RMNA competition season.

Engineering Intern

Plantaer

February 2026 - August 2026

- Led end-to-end hardware development of a solar-powered, off-grid smart irrigation system — from prototyping to field-ready deployment — using Raspberry Pi 4, solenoid valves, relay control logic, and a Python-Flask web server.
- \$20K customer pilot deliverable of a deployed solar-powered smart irrigation system on Governors Island, validating field performance in a real-world environment.

Data Science Fellow

CUNY Tech Prep

July 2025 - July 2026

- Selected for a competitive fellowship mentored by Amazon engineers; presented NeuroPBR, an end-to-end PBR material reconstruction system, at LinkedIn HQ.

AI/Software Intern

Quiztronics (RFCUNY)

June 2025 – September 2025

- Implemented a Retrieval-Augmented Generation pipeline with LangChain, JSON vector databases, and MySQL, improving accuracy of C++ tutoring responses by over 50%.

Robotics Research Assistant

CUNY

December 2024 - January 2026

- Built a ROS-based emotionally intelligent robot tutor integrating speech, vision, and gesture recognition with SLAM, enabling adaptive human-robot interaction for autistic children.
- Presented research findings on emotionally intelligent robot tutoring at the BARS Symposium to an audience of 100+ researchers and faculty.

PROJECTS

Bus Route & Neighborhood Risk Analysis • September 2025

- Built a Python dashboard leveraging live NYC MTA data to visualize bus route risk metrics; project was formally reviewed by MTA engineers for potential implementation and system optimization insights.

Warehouse Robot Simulation • September 2025

- Developed a ROS 2 warehouse robot system integrating SLAM and Nav2 for autonomous navigation and an AI-based dispatcher that optimized task allocation and collision avoidance.